

From Concern Detection to Promotion Control: A Protocol for Fail-Closed Governance of Autonomous Research Artifacts

Anonymous ACL submission

Abstract

Autonomous research systems increasingly produce complete-looking manuscripts and run bundles, yet review components can identify serious scientific defects without preventing a paper-ready decision. We formulate this gap as *promotion governance*: deciding whether an artifact bundle should be promoted, reviewed, downgraded, or blocked, while linking each blocking concern to inspectable evidence and a responsible repair stage. We introduce a manifest-driven evaluation protocol with clean controls, counterfactual fault variants, base-bundle-disjoint splits, blind manuscript-only requests, and metrics for false promotion, concern–acceptance conflict, clean-case retention, repair localization, traceability, and cost. A synthetic development suite with four base bundles and 40 cases validates the evaluator and exposes three missing integrity checks in an artifact audit. After those checks were added, the audit passed all development cases; an advisory-only ablation still accepted every defective case despite detecting its concerns. These results are instrument validation, not empirical evidence for general effectiveness: the suite is synthetic, unreviewed, and explicitly ineligible for paper-level claims. We specify an oracle-certified controlled experiment with held-out fault families and repeated real-model evaluation needed to test the central hypothesis without treating human adjudication as metric gold.

1 Introduction

Research-agent systems span literature review, experimentation, and report writing (Schmidgall et al., 2025). Capability benchmarks now evaluate replication workflows extending from paper understanding to experiment execution (Starace et al., 2025). Completion, however, is not equivalent to scientific readiness. BadScientist shows that plausible but unsound papers can receive favorable automatic reviews (Jiang et al., 2026), while counterfactual

evaluation finds that flaws in research logic have no significant effect on automatic reviews (Dycke and Gurevych, 2026).

We study the policy boundary between detecting a concern and allowing an artifact bundle to advance. A useful research governor must fail closed: a blocking concern cannot coexist with a paper-ready decision unless the concern is resolved or explicitly downgraded. It must also avoid the trivial solution of blocking everything. This motivates three questions: whether artifact-bound gates reduce false promotion, whether binding concerns to transitions prevents concern–acceptance conflict, and what clean-case, repair-localization, latency, and cost trade-offs follow.

The contribution at the present evidence level is a protocol and tested instrumentation, not a confirmed performance claim. We provide (i) a portable artifact-bundle task with explicit promotion and repair labels, (ii) leakage and independent-oracle gates, (iii) baselines that isolate detection from action binding, and (iv) a development failure analysis that routes audit misses back to the responsible workflow stages.

2 Related Work

Research-agent capability. Agent Laboratory organizes literature review, experimentation, and reporting within a multi-agent research workflow (Schmidgall et al., 2025). PaperBench evaluates replication of published machine-learning research from paper understanding through executed artifacts (Starace et al., 2025). These systems motivate artifact-level evaluation, but their primary object is research capability rather than the consistency of a promotion decision with detected blockers.

Review and claim verification. BadScientist operationalizes concern–acceptance conflict for unsound generated papers (Jiang et al., 2026). Dycke and Gurevych use targeted edits to create flawed

research logic and compare automatic reviews (Dycke and Gurevyh, 2026). CLAIM-BENCH evaluates claim extraction, evidence extraction, and claim–evidence link validation in AI papers (Javaji et al., 2025). These dimensions motivate a separate evaluation question: whether detected concerns constrain promotion decisions, bind artifact evidence, and identify the responsible repair stage.

Artifact assurance and recovery. ARA reconstructs scientific workflows for reproducibility assessment (Riehl et al., 2026). REPRO-Bench evaluates agents’ reproducibility assessments using paper PDFs and reproduction packages (Hu et al., 2025). SAGE routes experiment failures to structured intervention levels (Ma et al., 2026). The closest distinction is therefore not artifact inspection or failure attribution alone, but whether detected scientific blockers deterministically constrain paper-readiness promotion without over-blocking clean bundles.

3 Promotion Governance Protocol

3.1 Task and labels

Each case is a portable research-run artifact tree paired with a private case manifest. The evaluated system returns one decision from `promote`, `needs_review`, `downgrade`, or `block`; zero or more concerns with severity and evidence paths; and the workflow stages responsible for repair. Gold labels remain outside the visible artifact tree.

Every suite declares an evidence class, evaluation regime, claim ceiling, adjudication status, mutation-isolation status, and paper-claim eligibility. The controlled regime derives gold from a frozen fault registry and requires an independently implemented oracle to replay each mutation and verify the resulting artifact hash. Its claim ceiling is limited to registered fault families and it cannot claim external validation. A separate naturalistic regime retains independent human adjudication and external source provenance. All variants derived from one base bundle remain in the same split, and source content hashes detect renamed duplicates across splits.

3.2 Fault families and controls

The protocol requires clean positive, null, and negative controls together with variants covering missing comparison evidence, repeated-run provenance, hidden failed execution, declared-versus-executed

budget mismatch, result–figure conflict, claim–evidence conflict, citation support mismatch, stale persisted state, and unsupported claim strength. Each mutation records before and after hashes. For controlled evaluation, the implementation assigns entire fault families to development or test, derives case gold from the registry, and verifies every clean and mutated artifact by independent replay. Certification binds registry, gold, split, oracle-report, and development-suite hashes; any mismatch quarantines the suite. Evaluated systems receive only the artifact view, and blind manuscript requests omit case, family, path, oracle, and gold metadata. Naturalistic evaluation can additionally use separate mutation auditors and promotion-label reviewers.

3.3 Comparison conditions

We define four deterministic conditions and one provider condition. *Always promote* trusts the incoming ready state. *A presence checklist* verifies required files without semantic consistency checks. An *advisory artifact audit* reports the same concerns, evidence, and repair owners as the full audit but does not bind them to the decision. The *fail-closed artifact audit* applies the binding policy. Finally, a blind *manuscript-only reviewer* receives only paper text under opaque request identifiers; case labels, mutations, paths, and gold data remain private.

3.4 Metrics and inference

Primary metrics are false paper-ready promotion and concern–acceptance conflict. We also report four-way decision macro-F1, clean-case promotion, blocker precision/recall/F1, exact repair-owner match, trace coverage, latency, and provider cost. Every system trial must cover every case. Pairwise effects are computed per case, with 5,000-replicate bootstrap intervals clustered by base bundle and exact paired sign tests over base-bundle effects. Synthetic development suites remain exploratory regardless of numeric performance.

4 Implementation and Development Validation

The protocol is implemented as manifest-driven builders, runners, scorers, and blind request import/export commands. Paths are confined to suite roots; symbolic links, duplicate cases, hash mismatches, split leakage, and incomplete trial coverage are validation failures. Failure analysis emits

Condition	False	Conflict	Clean
Always promote	1.00	–	1.00
Presence checklist	1.00	–	1.00
Advisory audit	1.00	1.00	1.00
Fail-closed audit	.00	.00	1.00

Table 1: Synthetic development results over four base bundles and 40 cases. Columns report false-promotion, concern–acceptance conflict, and clean-case promotion rates. The labels were generated for instrument debugging and were not independently adjudicated; the table is not confirmatory evidence.

review diagnostics and node-strengthening recommendations with explicit recheck conditions.

The development corpus contains four synthetic base bundles, one clean control and nine counterfactual variants per bundle, for 40 cases. It is marked as synthetic development evidence with unreviewed adjudication and no paper-claim eligibility. Table 1 therefore tests instrumentation only.

Before strengthening, the fail-closed audit reached .70 decision accuracy and .333 false-promotion rate, missing repeated-run provenance, executed-budget, and stale-state faults. The failure analyzer routed the first two to experiment execution and the third to review. Added integrity checks were then evaluated on the same frozen suite. In the final development run, the fail-closed audit differed from the advisory ablation by .90 decision accuracy and -1.00 false promotion. The cluster bootstrap intervals are degenerate because each synthetic fault repeats identically across only four base bundles; the paired sign-test value is .125. This is direct evidence that the development sample is too small for inferential claims, despite final decision accuracy and blocker F1 of 1.00 on the generated cases.

5 Confirmatory Study and Readiness Gate

The controlled confirmatory study requires at least 720 held-out cases. Entire registered fault families are assigned to either development or test before outcomes are observed; base identities and source hashes are also disjoint. Every test base contributes one clean control and one case from each held-out test family. Certification independently replays all development and test mutations from their clean controls, verifies registry-derived gold, and binds the complete case set and split manifests by hash. A certified suite may become paper-claim eligible for registered fault families without human label

adjudication, but it remains explicitly unvalidated on naturally occurring defects.

The manuscript-only condition requires three preserved real-model trials per case. Remote API trials retain hashed provider receipts; local trials retain an exact model-artifact digest and hash-bound runtime receipts. Deterministic baselines and ablations use versioned, hash-bound system-run manifests. At least one post-repair rerun per original fault case must measure recovery and clean-control regression. Raw requests, responses, decisions, failures, manifests, hashes, latency, and cost must be retained. The confirmatory gate checks held-out family coverage for every system and trial, recomputes metrics from predictions, and cannot raise the claim ceiling beyond the registered families.

The central hypothesis is that fail-closed governance lowers false promotion by at least 20 absolute percentage points relative to the presence checklist while retaining at least 90 percent clean-case promotion. The study is stopped or downgraded if the full policy blocks more than 10 percent of clean bundles, if the checklist is within five points on both primary safety and clean-case metrics, if mutations leak labels, if oracle replay or split isolation fails, or if required real-model and recovery evidence is incomplete.

6 Limitations and Ethical Considerations

Counterfactual faults may be easier than naturally occurring research defects, and deterministic checks can overfit a known artifact schema. A promotion gate does not establish scientific truth; it enforces consistency between available evidence, stated concerns, and allowed transitions. Provider-based reviewers introduce model, prompt, and temporal dependence. Public artifact bundles must exclude credentials, private paths, personal data, and proprietary research content. False promotion can amplify unsound claims, while false blocking can suppress valid null or negative results; both errors must remain visible.

7 Conclusion

Promotion governance treats paper readiness as an evidence-bound transition rather than a by-product of workflow completion. The implementation and development suite show that this distinction is measurable and that detected failures can drive targeted audit strengthening. They do not yet show that the policy generalizes. An oracle-certified con-

trolled suite, repeated real-model review, complete recovery evidence, and a frozen confirmatory run are required before advancing beyond a research memo. Naturalistic generalization remains a separate claim that would require independently curated external evidence.

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